

Element G

The construction of our design began with the basic structure using pvc pipe. We added multiple substructures such as the mounting structure, the extending structure and the flexible structure for the hockey stick. All of these combine to give us our solution.

The mounting system is made up of a pvc pipe and velcro. This subsystem is placed on both sides of the design so that the stakeholders are able to attach the design to a wheelchair. It works by aligning the pvc to the wheelchair arms and using two velcro straps on both sides to wrap around the arm so that it is secure to the chair.

The extending system was not originally part of the design but was later included so that our solution can be used on any sized wheelchair. It consists of a large pvc pipe, a smaller pvc pipe and a custom 3D printed adapter. The way it works is that the smaller pipe will fit inside the larger pipe so that it can slide back and forth, the adapter is glued on to the big pipe so that the space in between the pipes is limited in order for it to be fixed to any given position but still move if physically pulled.

Lastly the flexible structure which is what is used so that the hockey stick has a mostly free range of motion. It simply consists of a pvc T connector.

The hockey stick is too short for the design, so we're going to extend the hockey stick length to make it fit our project. The hockey stick is placed inside of the pvc connector so that the hockey stick is held in place and be able to move if you use your hands to move it.